

PCA and Hierarchical Cluster Modelling

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Revisiting Dependent Variables

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Sociology and classification

“In its conceptual purity, this mental construct cannot be found empirically anywhere in reality. It is a utopia.”

– Max Weber, *The “Objectivity” of Knowledge in Social Science and Social Policy* (1904)

- Comparative sociology and political science classify in order to understand.
- Welfare regimes, varieties of capitalism, party families – typologies in the Weberian tradition.
- PCA and clustering do the same thing from the data side: variation among cases suggests which dimensions and which groupings matter.

What we will do

1. Understand PCA, hierarchical clustering, and k -means as exploratory tools.
2. Show how PCA relates to clustering to produce institutional typologies.
3. Example: welfare-state financing across OECD countries.
4. Hands-on in R: ALMP / labour-market policy mix in Europe.

Branches of machine learning

Setting	Data	Methods	Example
Unsupervised learning	No outcome labels	Clustering, principal component analysis mixture models	Group welfare states with similar profiles
Supervised learning	Labeled data: outcomes known	Decision trees, random forests, regularised regression, neural nets	Predict unemployment duration from demographics, job history, and local labor market factors
Reinforcement learning	Learns from feedback through interaction	Multi-armed bandit, Q-learning	Learn which job support actions speed up re-employment

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The unsupervised machine learning toolkit

Method	What it does	Output
PCA	Project to low-dim continuous space	Scores, loadings
Hierarchical clustering	Build a tree of nested groupings	Dendrogram
<i>k</i> -means	Partition into <i>k</i> disjoint groups	Cluster labels
MCA / MFA	PCA for categorical / mixed data	Scores, loadings
Latent class	Probabilistic clustering	Posterior probabilities

All are unsupervised: no outcome variable, the structure is inferred from the data.

PCA recap

PCA finds orthogonal linear combinations of variables that successively maximise variance.

$$\mathbf{X}_{n \times p} \longrightarrow \mathbf{Z}_{n \times p} = \mathbf{X}\mathbf{V}, \quad \mathbf{V}^{\top}\mathbf{V} = \mathbf{I}$$

- Columns of $\mathbf{V}$ are eigenvectors of the covariance matrix of standardised $\mathbf{X}$.
- Eigenvalues λ_k give the variance explained by component k .
- For typology research we usually keep the first 2–4 components.

We can combine PCA with clustering

Welfare-state typologies share three properties that make this pairing useful:

- The type is latent.
- There are many indicators, often correlated.
- There is no ground-truth label, so we want the data to suggest groupings rather than impose them.

PCA reduces noise and puts all indicators on the same scale. Clustering then groups countries in this reduced space.

Roadmap

1. Intro
2. From PCA to clustering – methods
3. Example: welfare-state financing
4. Hands-on: ALMP policy mix in R
5. Wrap-up and limitations

Bias and variance in unsupervised settings

The classical bias-variance decomposition is a supervised concept – it presumes a true target. Without labels, we use it as an analogue for the tuning choices ahead.

- PCA, number of components q . Few q projects away real signal (bias). Many q admits noise components that do not replicate (variance).
- Clustering, number of clusters k . Small k lumps unlike countries (bias). Large k produces small clusters that hinge on individual observations (variance).
- Linkage. Ward's biases toward spherical, balanced clusters with low resampling variance. Single linkage does not constrain cluster shape but is unstable across resamples.

We check how each choice holds up by resampling.

The combined pipeline

1. Standardise variables.
2. Run PCA, keep first q components ($q \ll p$).
3. Cluster on the q -dimensional scores.

Why PCA before clustering:

- Filters measurement noise and redundancy.
- Equalises geometric scale across indicators.
- Makes Euclidean distance interpretable.

Distance metrics

Almost all clustering rests on a distance metric.

- Euclidean – $d_{ij} = \sqrt{\sum_k (x_{ik} - x_{jk})^2}$. Sensitive to scale.
- Manhattan – robust to outliers.
- Mahalanobis – accounts for covariance; rarely used for typologies.

With standardised PC scores, Euclidean is the natural choice.

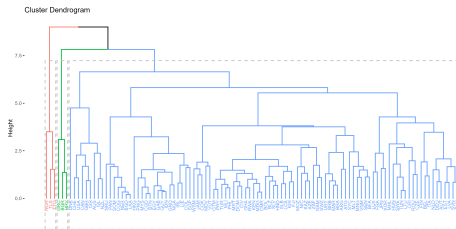
Hierarchical clustering – linkage methods

At each step, merge the two closest clusters. “Closest” depends on linkage:

- Single – distance of nearest pair. Tends to chain.
- Complete – distance of farthest pair. Compact, balanced clusters. (Used in the financing demo.)
- Average – mean pairwise distance.
- Ward's – merge that minimises within-cluster variance increase. Most common in welfare-state research.

Reading a dendrogram

- Vertical axis: merge distance.
- A horizontal cut at height h defines a partition.
- Lower the cut, more clusters.
- Big vertical gaps suggest natural cut points.



Validation

- Bootstrap stability – resample countries, re-cluster, count co-membership rates.
- Sensitivity – different linkage, different k , different variable set.
- External benchmarking – compare to Esping-Andersen, Ferragina, Soler-Buades *et al.*
- Face validity – do the cluster means tell a coherent substantive story?

Interpretation pitfalls

- The algorithm always returns a partition, even on data with no real structure.
- Axis rotation is arbitrary – PC1 vs PC2 has no causal meaning; signs can flip across software.
- Sample composition drives results – adding 30 low-income countries reshapes every loading.
- Compositional traps – if variables sum to a fixed total, correlations are mechanical.
- Missing-data choices change which countries are even in the analysis.

The full workflow in R

```
# 1. standardise
X <- scale(raw_data)

# 2. PCA
pca <- prcomp(X)
scores <- pca$x[, 1:3]    # keep first 3 PCs

# 3. hierarchical clustering on PC scores
d <- dist(scores)
hc <- hclust(d, method = "complete")
plot(hc)
clusters <- cutree(hc, k = 6)
```

Why financing

- Welfare-state typologies usually classify countries on spending: generosity, decommodification, social investment.
- Financing – who pays, through which instrument – is the other side. It shapes:
 - political coalitions for retrenchment and expansion,
 - distributional incidence,
 - adjustment capacity in crises.
- Yet financing is rarely treated as the dependent variable.

Data

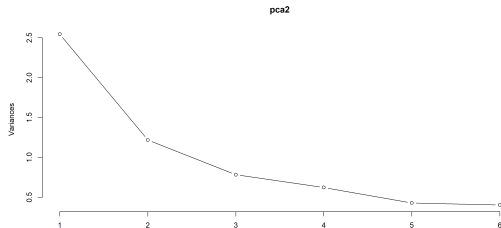
- Source: OECD Revenue Statistics, tax revenue by category, % of GDP.
- Coverage: about 100 countries (the analysis goes beyond the OECD core).
- Snapshots at 1990, 2002, 2012, 2022.
- Variables, after dropping the redundant total tax:
 - personal income tax (PIT)
 - corporate income tax (CIT)
 - social security contributions (SSC)
 - property taxes
 - sales / VAT (taxes on goods and services)
 - use of goods (excises, vehicle, environmental)

Standardising and dropping totals

- Tax categories have very different scales: SSC up to about 15% of GDP, property usually below 2%. Without scaling, PCA is dominated by the largest-variance variables.
- Total tax is mechanically the sum of categories. Including it inflates PC1 and obscures the financing-mix story.
- I standardise to mean 0, variance 1, and drop totals.

PCA – variance explained

- PC1: about 45% of variance.
- PC2: about 20%.
- Cumulative through PC2: about 65%.
- Sharp drop after PC3 – a two-component picture is sensible.



PCA – loadings

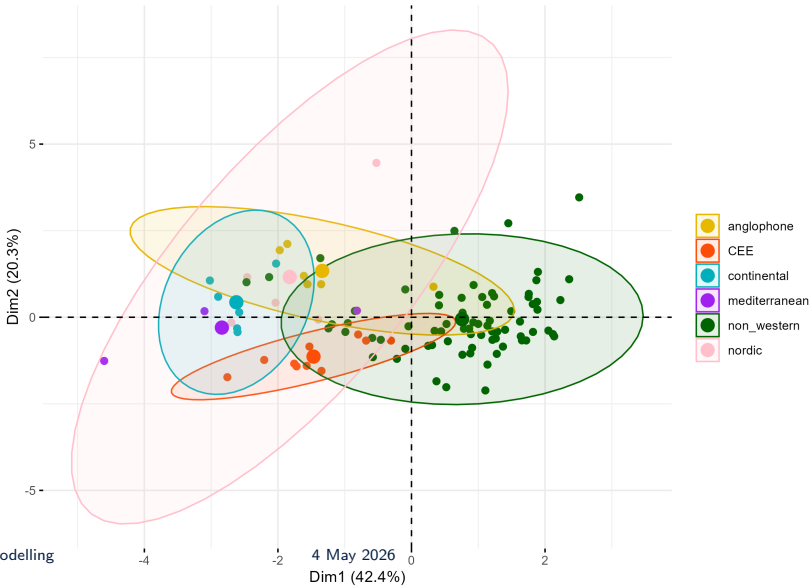
Variable	PC1	PC2
Use of goods	−0.51	−0.10
Social security contributions	−0.48	−0.09
Personal income tax	−0.46	+0.29
Property tax	−0.44	+0.36
Sales / goods and services	−0.33	−0.51
Corporate income tax	+0.04	+0.71

PC1 picks up the size of the tax state – negative scores for high-tax countries.

PC2 separates corporate-tax-heavy from consumption-tax-heavy financing.

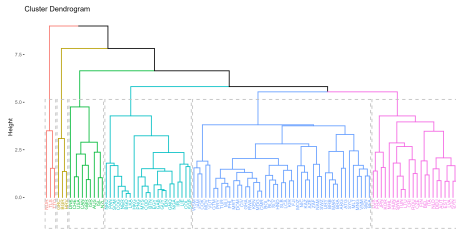
Countries in PC1-PC2 space

Individuals - PCA



Hierarchical clustering on PC scores

- Complete linkage on Euclidean distance.
- Big jumps in merge height suggest natural cuts.
- Headline cut at $k = 6$. We will look at $k = 4$ and $k = 8$ for robustness.



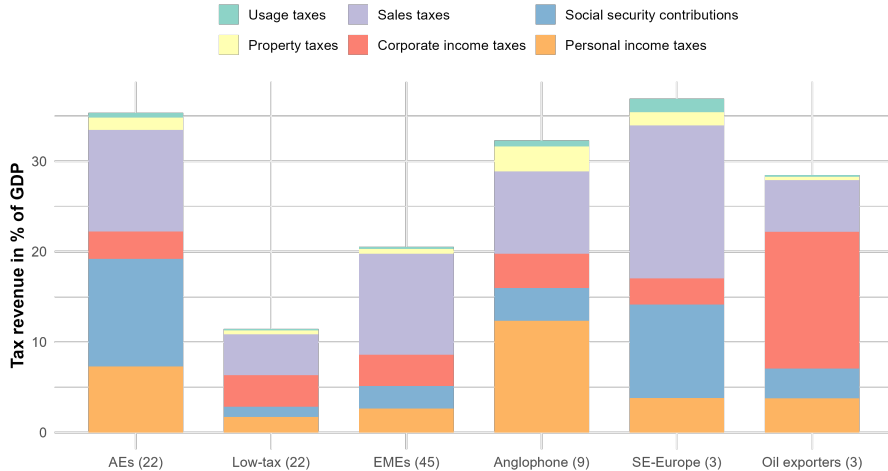
Six clusters, 2022

Cluster	Members (illustrative)
Advanced economies (22)	DEU, AUT, BEL, FRA, ITA, NLD, SWE, FIN, JPN, KOR, ...
Anglophone + DNK (9)	USA, GBR, AUS, NZL, CAN, CHE, ISL, ISR, DNK
SE-Europe (3)	GRC, HRV, BGR
Oil exporters (3)	NOR, TTO, TLS
Emerging markets (45)	BRA, IDN, EGY, MYS, ...
Low-tax (22)	EST, LVA, BGD, KEN, ...

Denmark groups with the Anglophone bloc. Both fund welfare mainly through personal income tax, rather than through social security contributions as in continental Europe.

Cluster means

Average tax revenues across country clusters



Robustness across k

- $k = 4$ – the Anglophone bloc collapses into AEs; oil exporters survive as their own cluster (driven by CIT).
- $k = 6$ – headline solution.
- $k = 8$ – AEs split into Continental vs Nordic + East Asian sub-types.
- Across all cuts, the Anglophone vs Continental divide is preserved.

Comparison to Esping-Andersen

- A clear Anglophone (liberal) cluster shows up, along with a distinct Continental cluster anchored by SSC.
- The Nordic group, however, does not hold together on the financing side. Sweden and Finland sit with Continental; Denmark with the Anglophone.
- Esping-Andersen typologises on spending structure (decommodification). On the financing side the country groupings come out different.

Change over time, 1990–2022

- PC1 (size of tax state) compresses over time: low-tax countries climb, high-tax countries plateau.
- PC2 (corporate vs consumption) spreads. CIT becomes more variable across countries; consumption taxes converge upward.
- Cluster boundaries are remarkably stable. The Anglophone vs Continental cut survives all four decades.

Take-aways from the demo

1. Financing structure has two main dimensions: the level of taxation, and the mix of instruments.
2. The financing typology produces different country groupings than spending-side typologies, even where some clusters look familiar.
3. The structure is stable across decades, which makes it a candidate dependent variable in panel research.
4. Method choices (standardisation, drop totals, linkage) matter and need to be reported.

What you will do

Run a full PCA + HC + k -means pipeline on the European ALMP policy mix.

- Data: OECD Labour Market Programmes database (DSD_LMP), 28 European countries, 2018–2022 average. Pulled via SDMX API.
- Variables: spending shares across PES, training, employment incentives, supported employment, direct job creation, start-up incentives.
- Files in the shared folder:
 - 00_fetch_oecd_lmp.R – pulls and cleans the data
 - 01_almp_pca_hc.R – the analysis you will run
 - almp_data.csv – pre-cleaned data
 - README.txt

Structure of the hands-on session

1. I walk through the script.
2. You run it end-to-end and inspect the outputs.
3. Bonus:
 - swap complete linkage for Ward's; does the typology change?
 - rerun without standardisation; which variables now dominate?
 - drop one country; is your cluster stable?

The script in one screen

```
library(tidyverse); library(factoextra)
df <- read_csv("almp_data.csv") |>
  column_to_rownames("country")

X <- scale(df)
pca <- prcomp(X)
fviz_eig(pca)
fviz_pca_biplot(pca, repel = TRUE)

scores <- pca$x[, 1:3]
hc <- hclust(dist(scores), method = "complete")
fviz_dend(hc, k = 4, rect = TRUE)

clusters <- cutree(hc, k = 4)
```

When the pipeline misleads

- Small N (below 30): PCs are noisy, clusters unstable.
- Mixed-type data (continuous + categorical): use MCA / MFA instead.
- Heavy outliers: one country can rotate the PCs.
- Compositional data: consider log-ratio transforms (Aitchison) before PCA.
- Time-varying institutions: consider PCA on changes, or pool with country fixed effects.

Beyond what we covered

- k -means – partition-based alternative to HC. Fix k upfront, iterate centroid assignments. Scales to large N .
- MCA – Multiple Correspondence Analysis, for categorical indicators.
- MFA – Multiple Factor Analysis, for variables organised in groups.
- Latent class analysis – probabilistic clustering with model selection.
- Fuzzy clustering – soft assignments when types blend (welfare research often does).

Thank you

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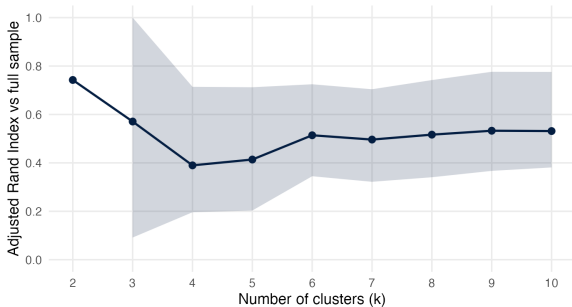
Slides, R script, and data in the shared workshop folder.

Key references

- Esping-Andersen 1990 – *Three Worlds of Welfare Capitalism*.
- Ferragina 2022 – compensatory vs employment-oriented policies, S&A.
- Hemerijck & Plavgo 2021 – social investment litmus test.
- Husson, Lê & Pagès 2017 – *Exploratory Multivariate Analysis by Example Using R*.
- Hastie, Tibshirani, Friedman 2009 – *Elements of Statistical Learning*, ch. 14.
- Soler-Buades, Natili & Biscuola, w.p. – minimum-income typology, 25 countries.

Cluster stability across k – financing data

- Subsample 80% of countries, refit PCA + complete-linkage HC.
- Adjusted Rand Index against the full-sample partition, 200 subsamples.
- ARI dips at $k = 4-5$ (partition is unsettled), then plateaus from $k = 6$.
- Narrower band at higher k – the partition holds together at this resolution.



200 subsamples, 80% of countries each. Welfare-state financing data, 2022. Complete linkage on PC1-PC3.